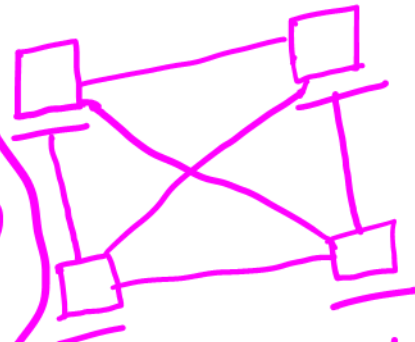


Forouzan
4th edition

Data Communication
&
Networking.



Network

Network

Components of a network:->

- (i) Sender
- (ii) Receiver
- (iii) Medium
- (iv) Message.
- (v) Protocol (set of Rules).

(i) Client $\xrightarrow{\text{Request}}$ Server
(ii) Server $\xrightarrow{\text{Response}}$ Client

Network Connecting
Devices

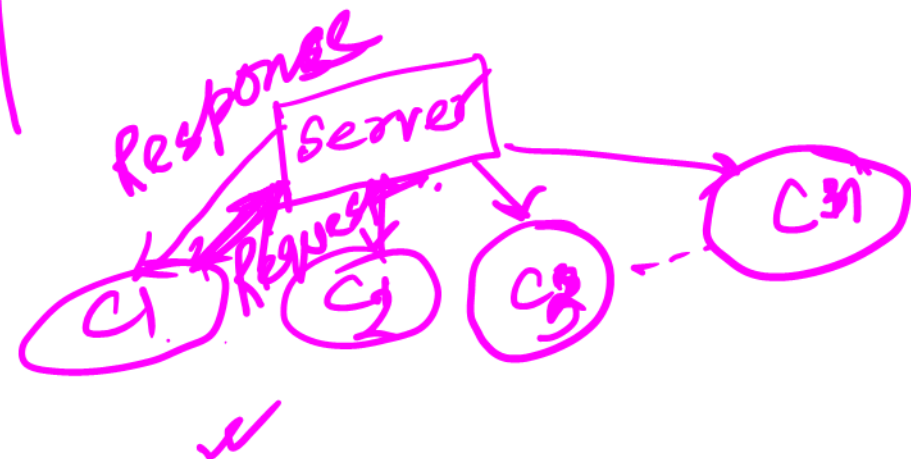
Adv. :->

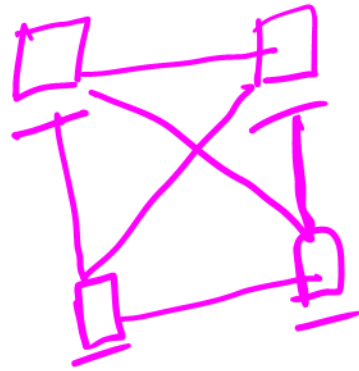
Disadv. :->

Type of Network

(i) Client-Server
Network

(ii) Peer to Peer Network





Peer to Peer Model.

Mode of Transmission.

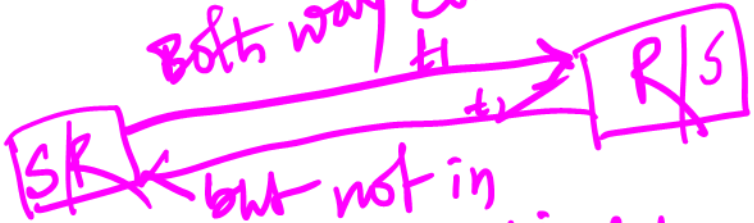
- ✓ (i) Simplex Mode
- ✓ (ii) Duplex Mode
 - Half Duplex
 - Full Duplex.

(i) Simplex Mode.



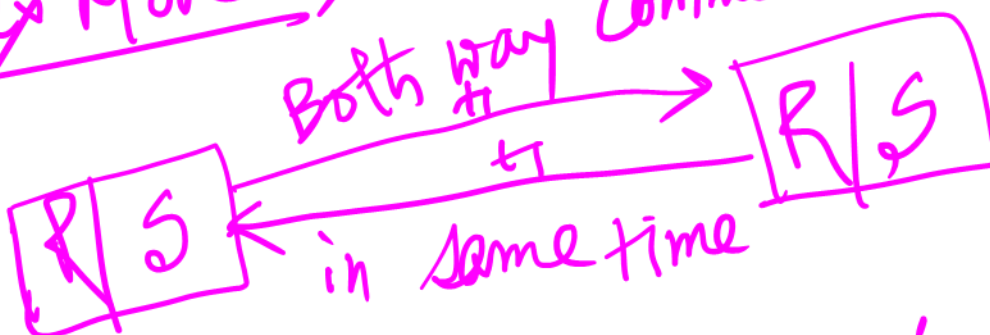
Ex: → Radio, Tr.
 $I \rightarrow R$

(ii) Half Duplex Mode! →

Both way communication
 t_1 t_2

 but not in same time.

Ex! → Walky Talky

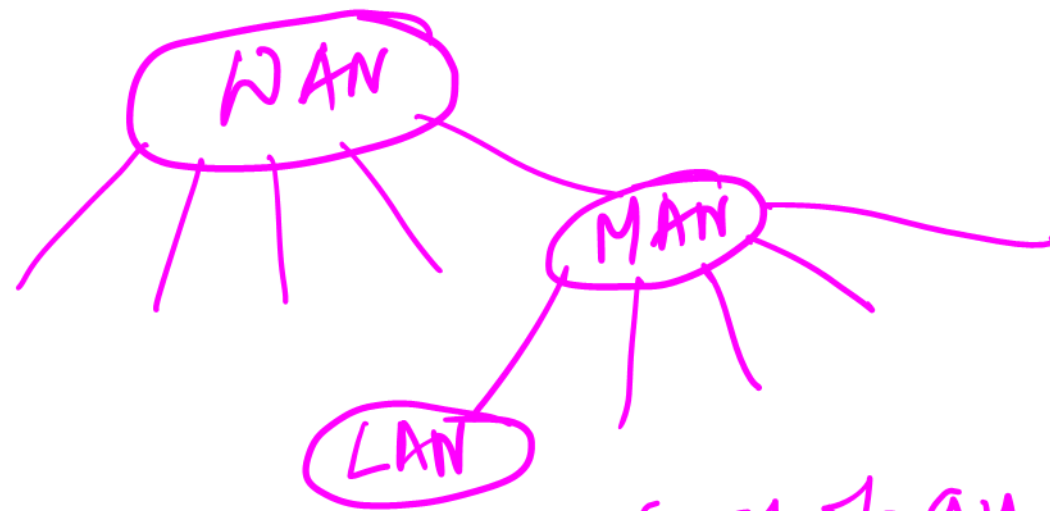
(iii) Full Duplex Mode! →

Both way communication
 t_1 t_2

 in same time

Ex! → Telephone

Type of Network

- PAN → (Personal Area Network)
- LAN → (Local Area Network)
- MAN → (Metropolitan Area Network)
- WAN → (Wide Area Network).

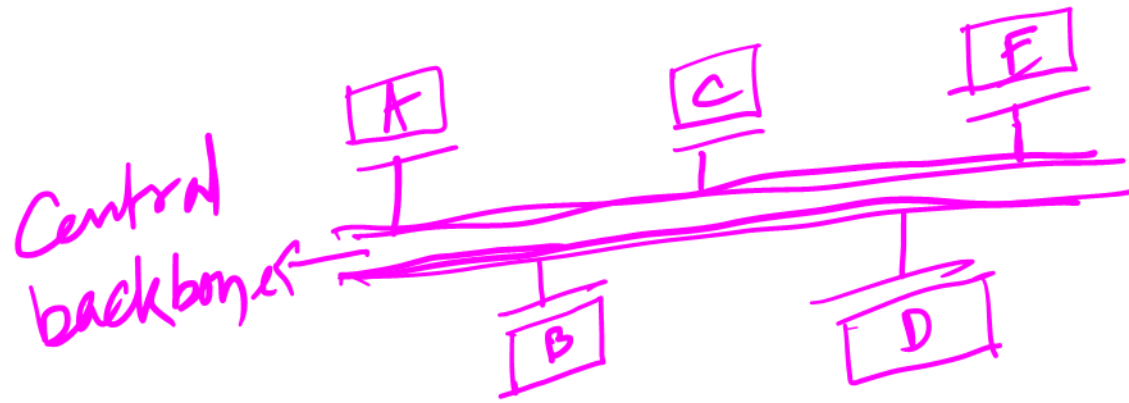


Physical structure of a network (Topology).
Graphical representation of a network is called topology.

Topology

- BUS Topology
- RING " "
- STAR " "
- Mesh " "
- Tree " "
- Hybrid " "

(i) BUS Topology :->



Adv.:

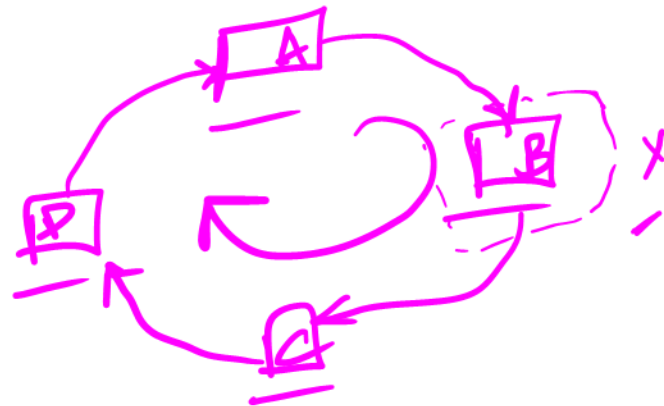
- (i) Cost efficient.
- (ii) Easy to install.
- (iii) Data transfer Easy. etc.

Dis-advantage.

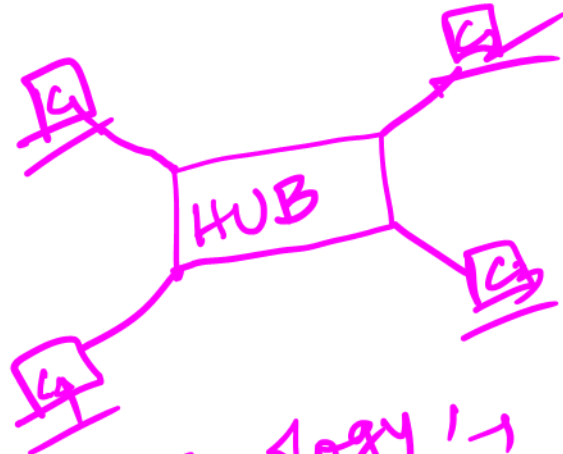
- (i) Data security.
- (ii) If central backbone collapse then the whole network will be fall.

(ii) RING Topology :->

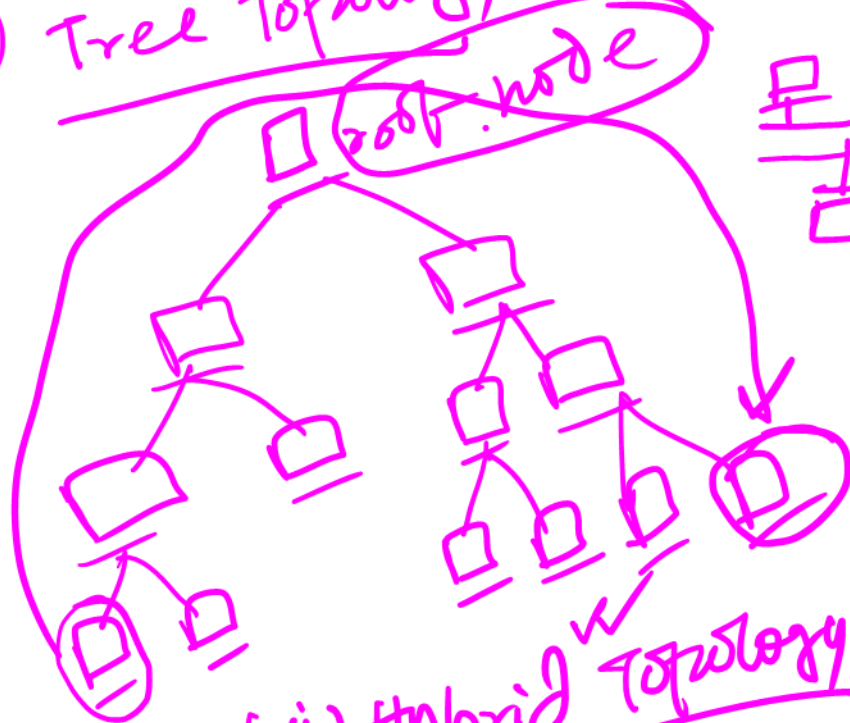
Token Ring
Token Bus.



(iii) STAR Topology! →

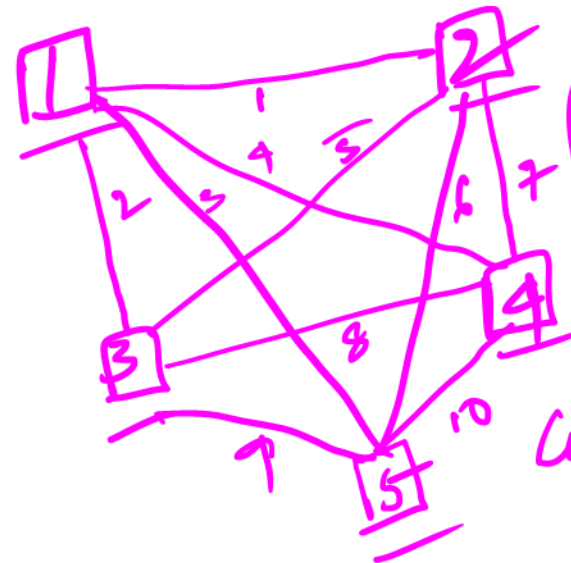


(v) Tree Topology! →



(vi) Hybrid Topology! →

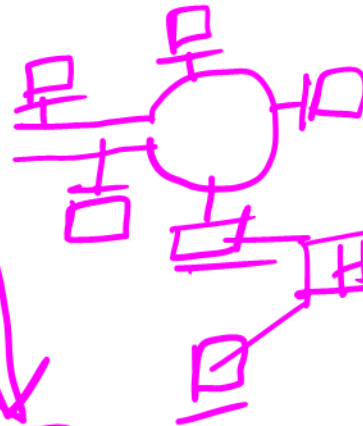
(iv) Mesh Topology



Cost.

For small network.

Complete Graph.



Total of Node → N .

Total of edge (Wire) → $\frac{N(N-1)}{2}$

$$\frac{5 \times 4}{2} = 10$$

BUS + RING + STAR

Hybrid Topology.